AutoML Model Audit Report

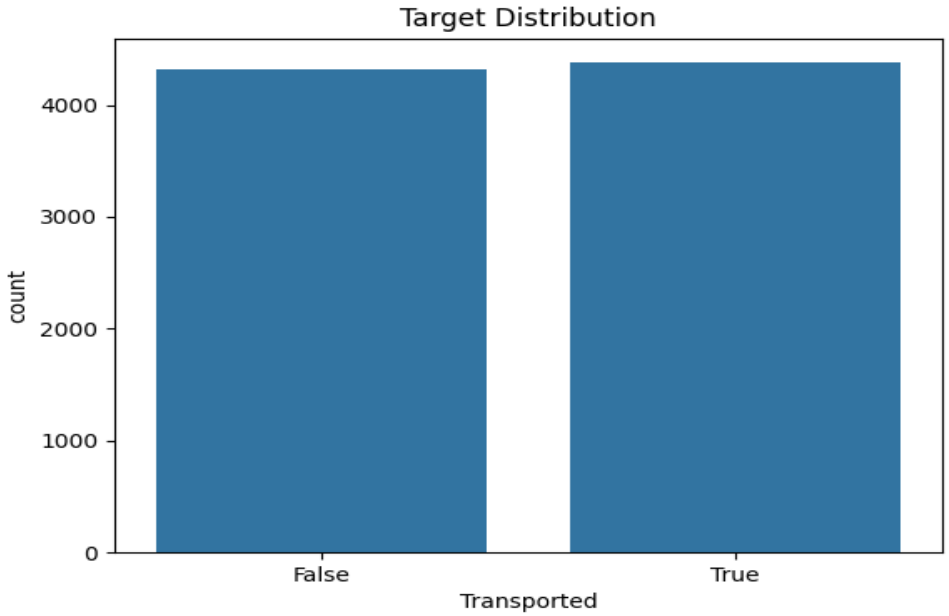
This document explains every step taken by the AutoML pipeline from raw dataset to final model selection.

Feature Statistics (Before Preprocessing)

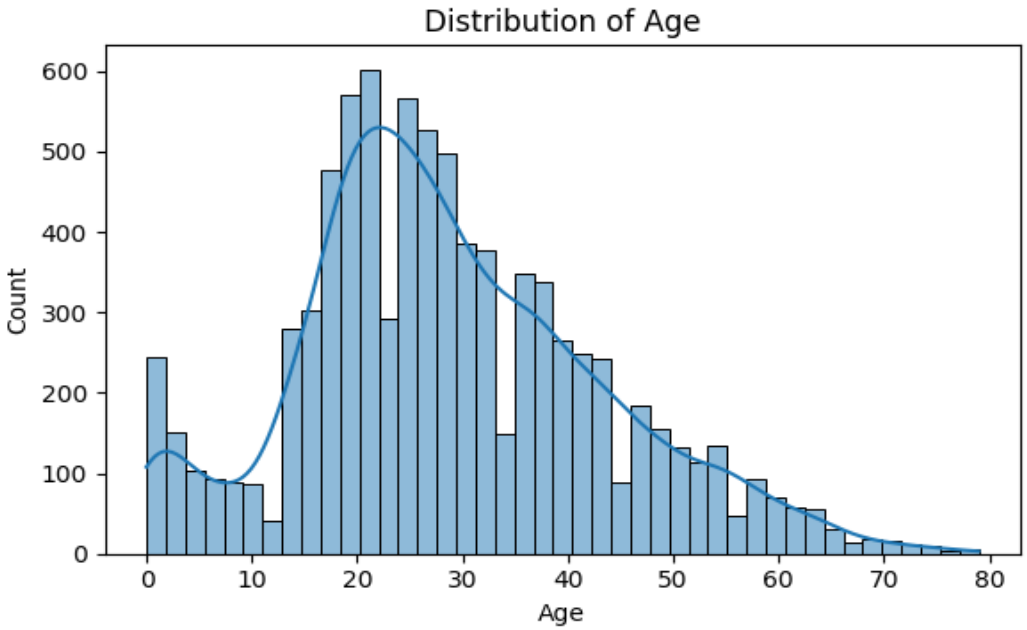
Feature	Type	Unique	Mean	Std	Min	Max
PassengerId	str	8693				
HomePlanet	str	3				
CryoSleep	object	2				
Cabin	str	6560				
Destination	str	3				
Age	float64	80	28.828	14.489	0.0	79.0
VIP	object	2				
RoomService	float64	1273	224.688	666.718	0.0	14327.0
FoodCourt	float64	1507	458.077	1611.489	0.0	29813.0
ShoppingMall	float64	1115	173.729	604.696	0.0	23492.0
Spa	float64	1327	311.139	1136.706	0.0	22408.0
VRDeck	float64	1306	304.855	1145.717	0.0	24133.0
Name	str	8473				
Transported	bool	2	0.504	0.5		

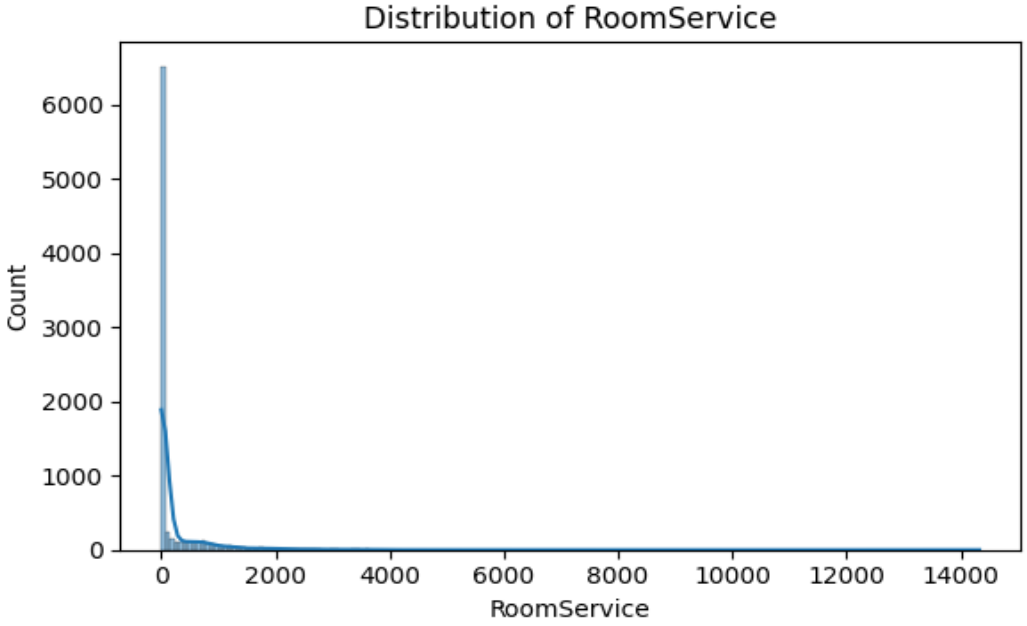
Dataset Overview

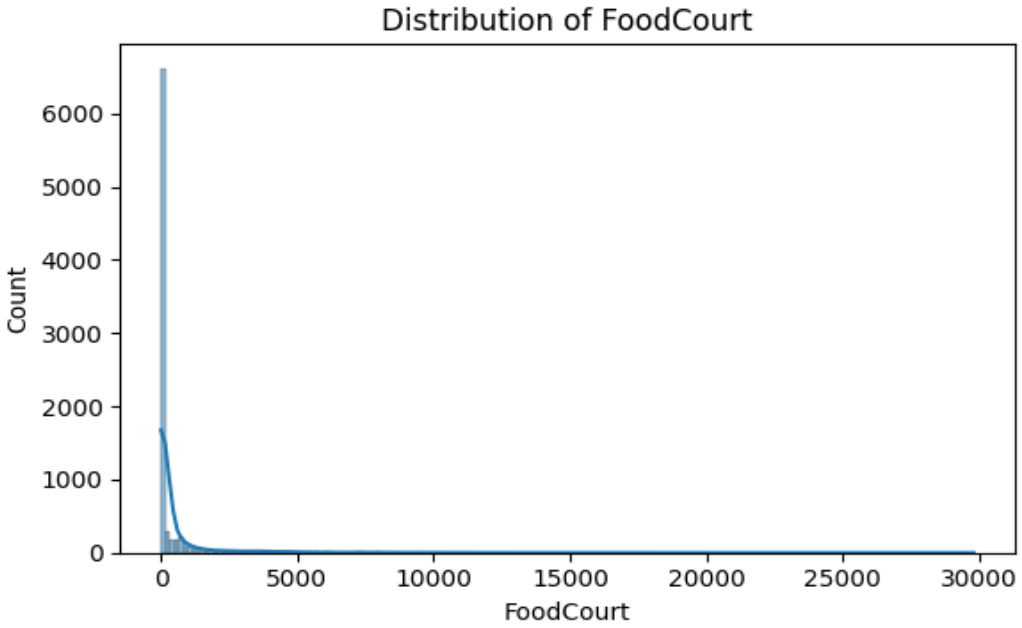
Rows: 8693 | Columns: 14

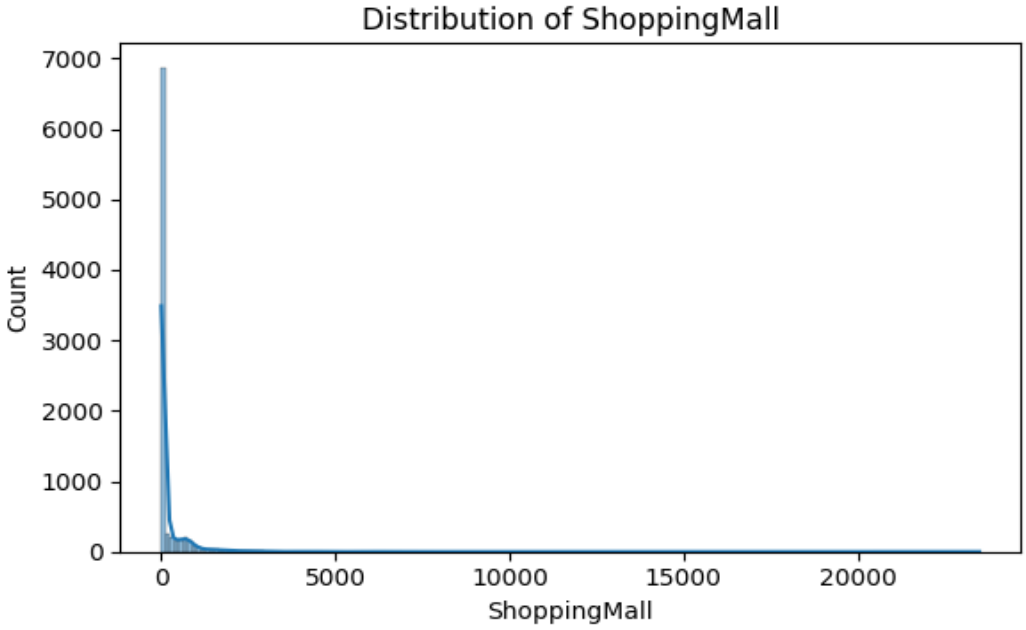


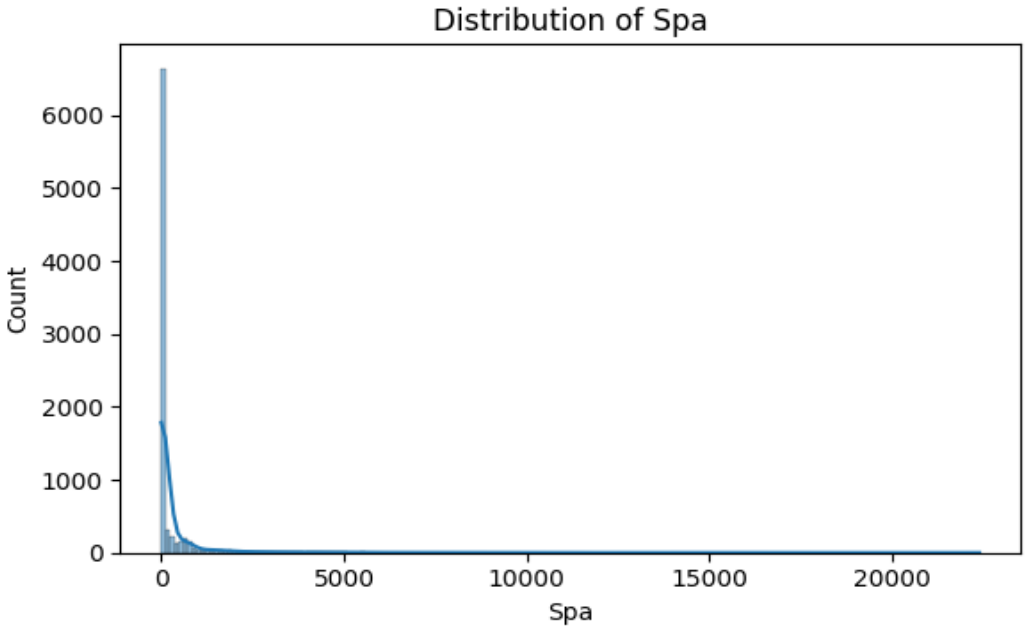
Exploratory Data Analysis

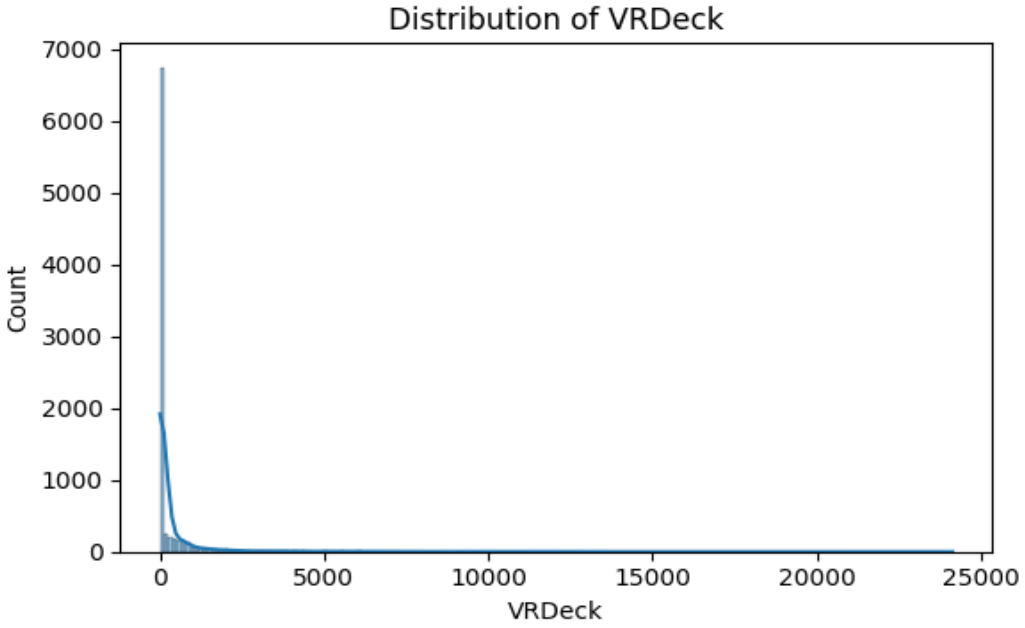


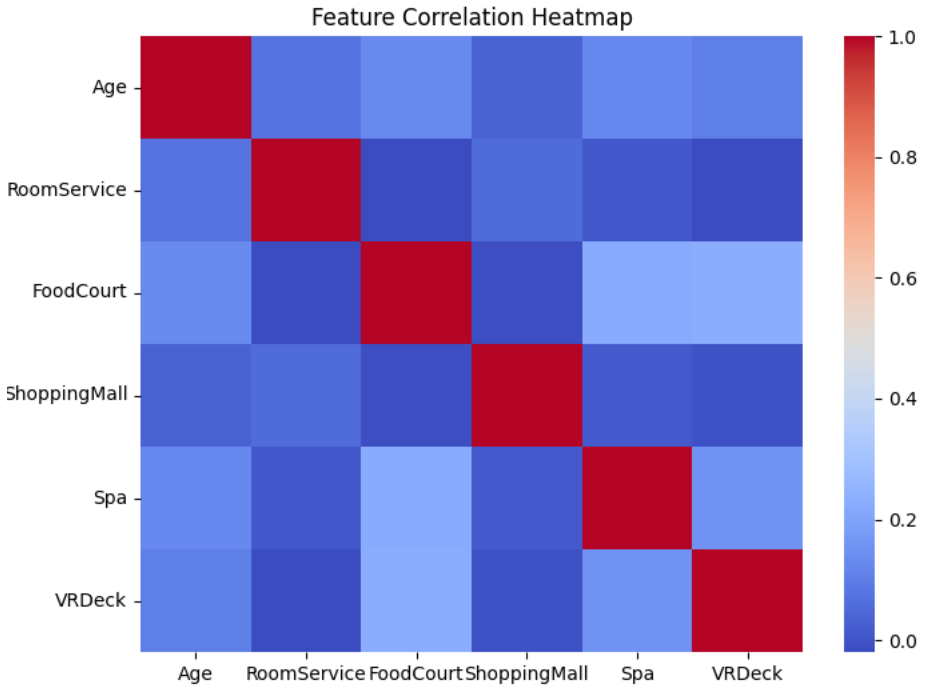



Distribution of FoodCourt
Count
FoodCourt
0
1000
2000
3000
4000
5000
6000
0
5000
10000
15000
20000
25000
30000

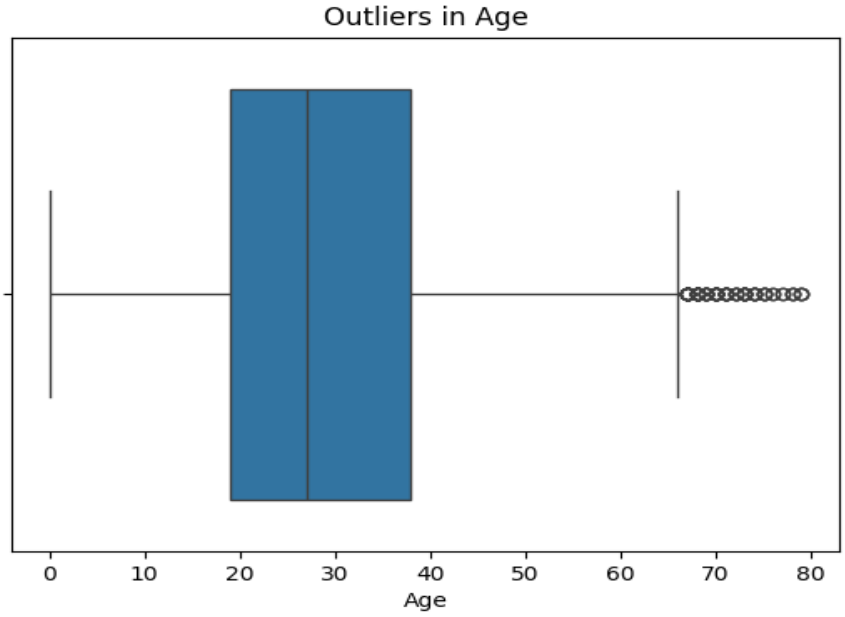

Distribution of ShoppingMall
Count
ShoppingMall
0
1000
2000
3000
4000
5000
6000
7000
0
5000
10000
15000
20000

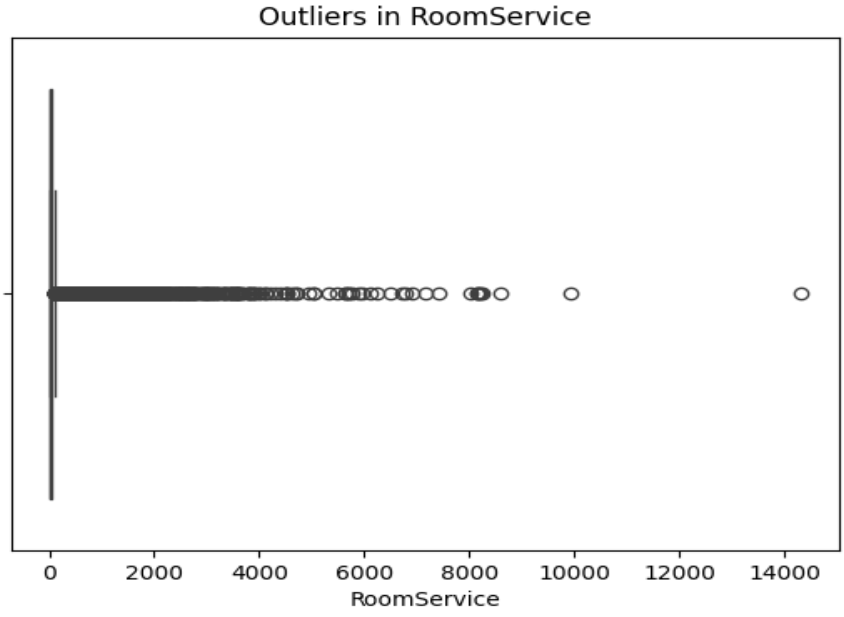

Distribution of Spa
Count
Spa
0
1000
2000
3000
4000
5000
6000
0
5000
10000
15000
20000

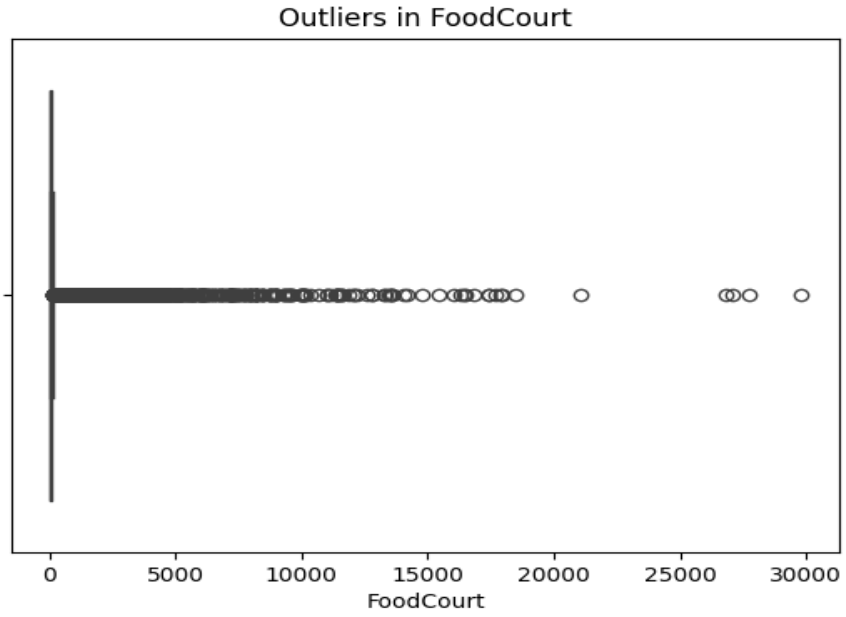

Distribution of VRDeck
Count
VRDeck
0
1000
2000
3000
4000
5000
6000
7000
0
5000
10000
15000
20000
25000

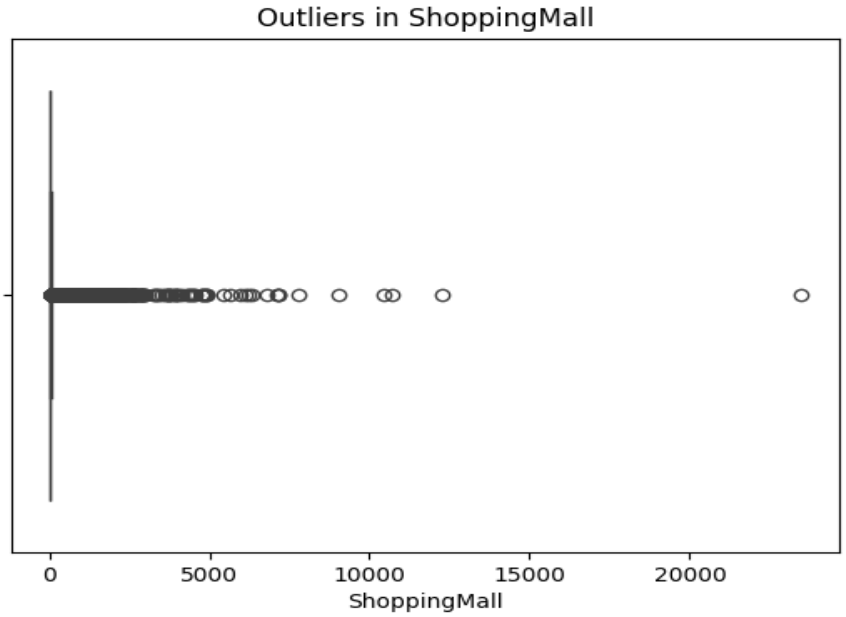

Feature Correlation Heatmap
Age
RoomService
FoodCourt
ShoppingMall
Spa
VRDeck
Age
RoomService
FoodCourt
ShoppingMall
Spa
VRDeck
1.0
0.8
0.6
0.4
0.2
0.0

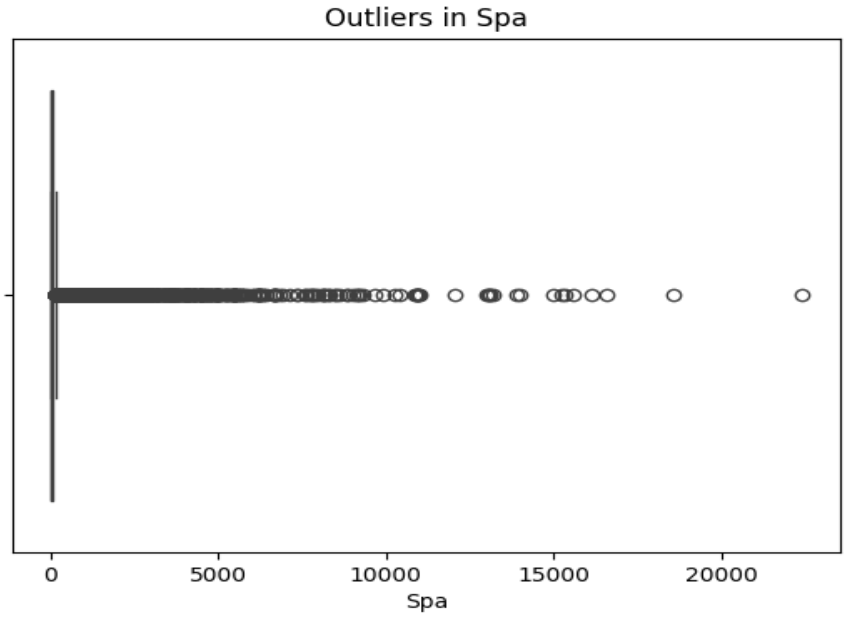
Outlier Detection

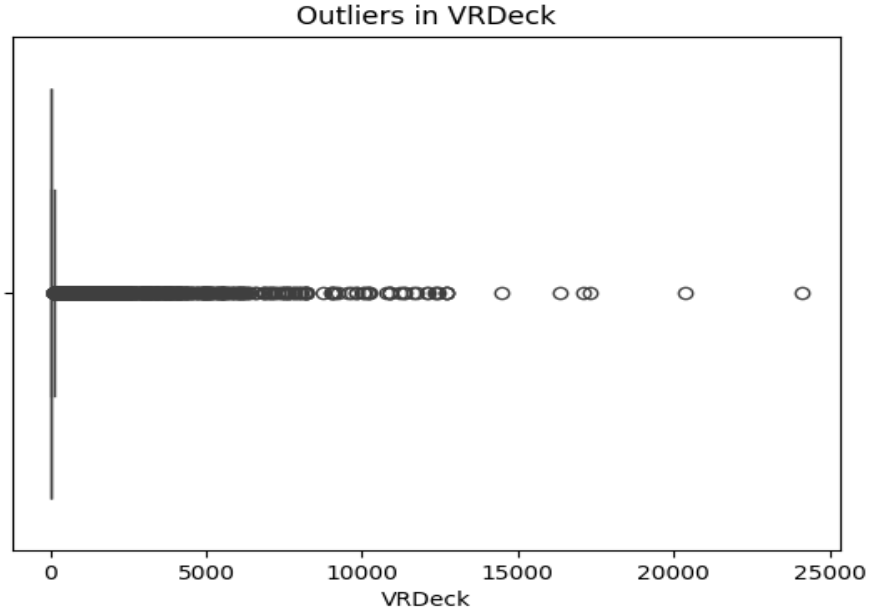




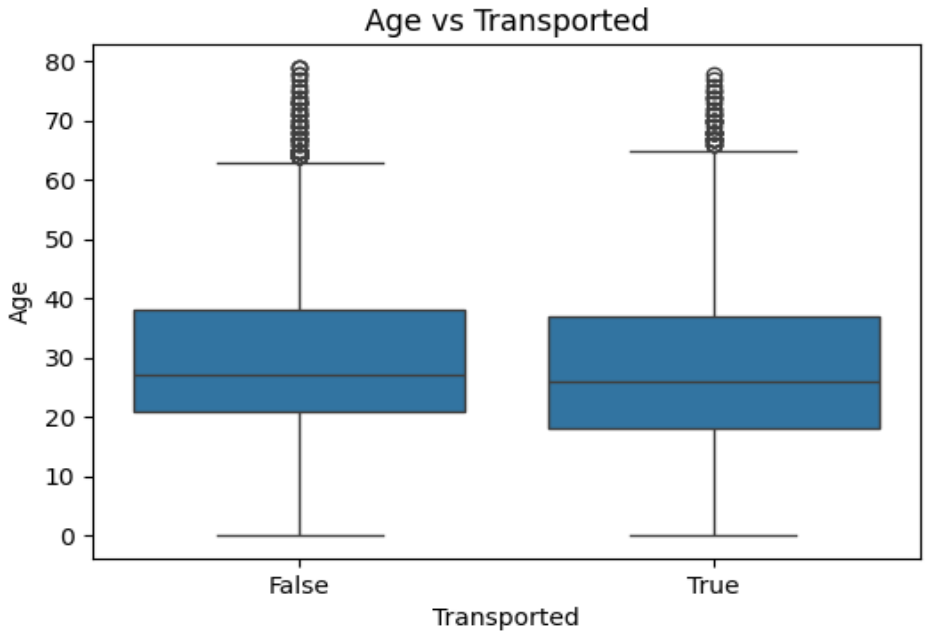

Outliers in FoodCourt
0
5000
10000
15000
20000
25000
30000
FoodCourt

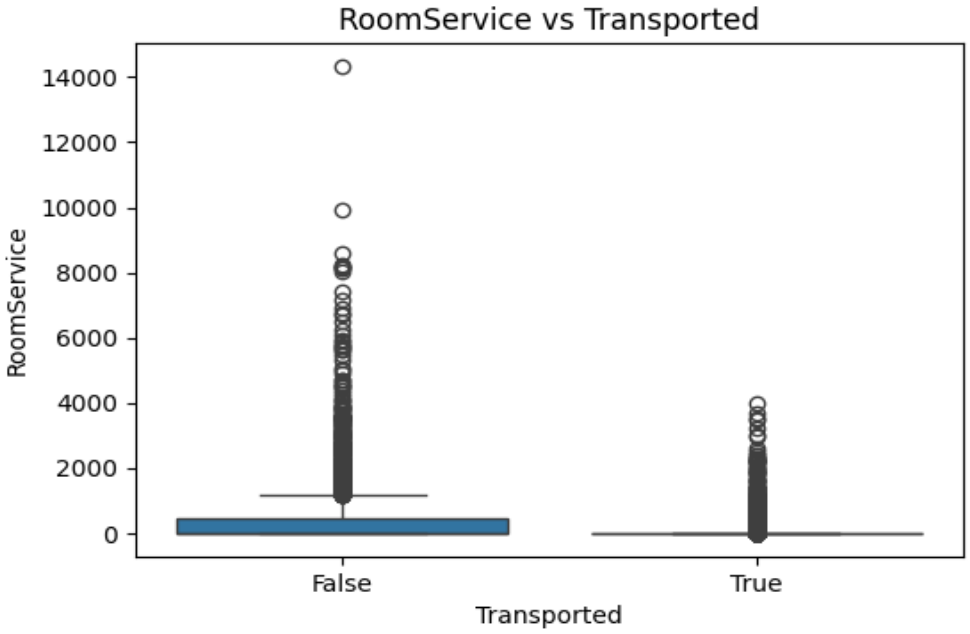

Outliers in ShoppingMall
0
5000
10000
15000
20000
ShoppingMall

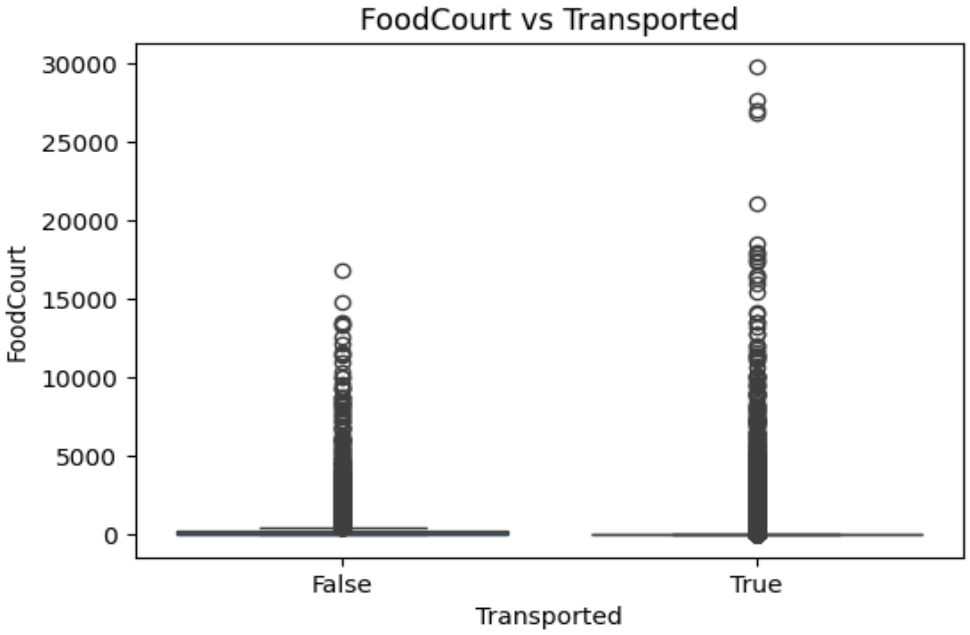

Outliers in Spa
0
5000
10000
15000
20000
Spa

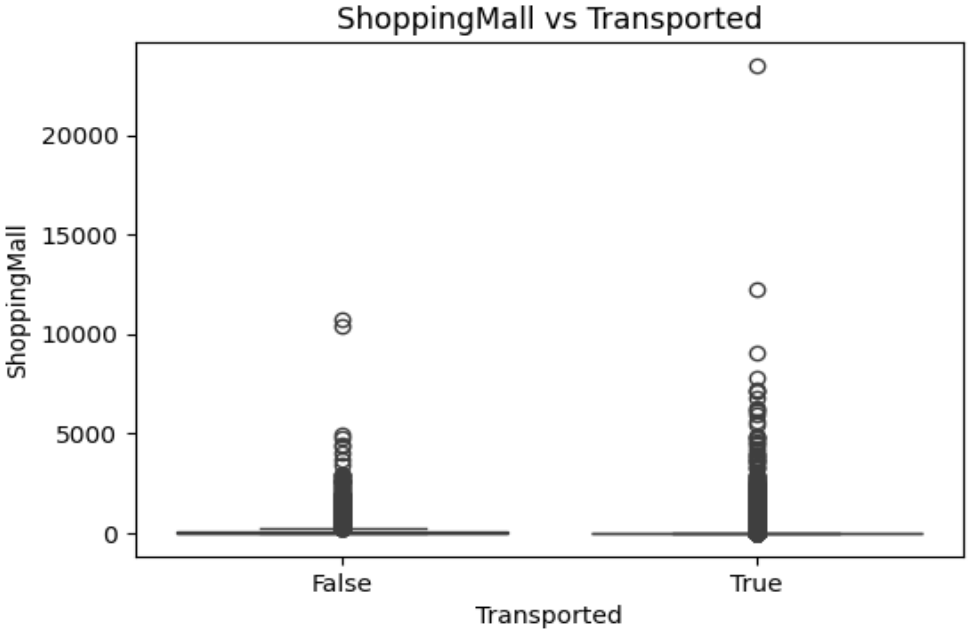

Outliers in VRDeck
0
5000
10000
15000
20000
25000
VRDeck

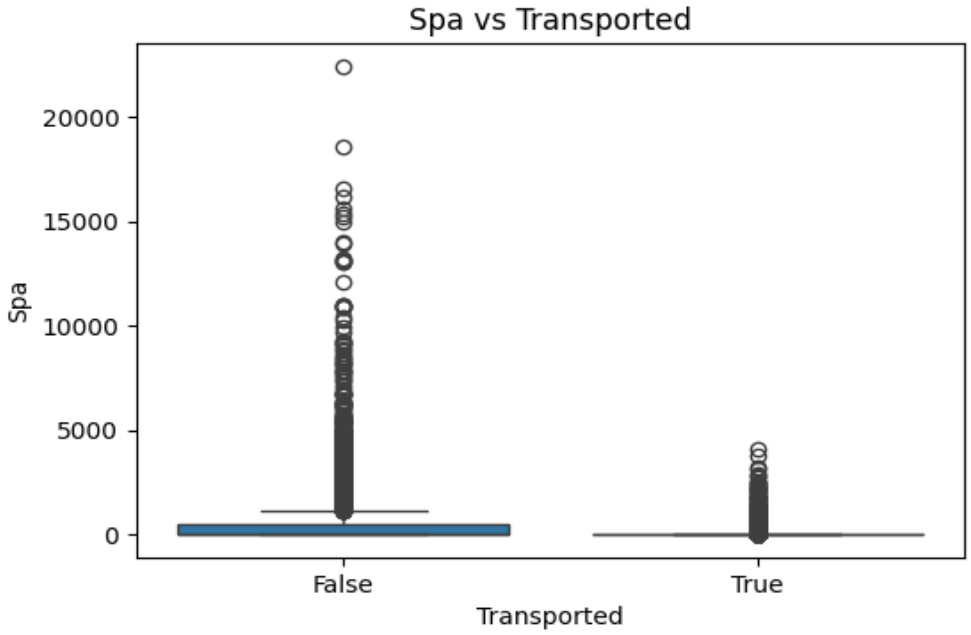
Feature Impact on Target

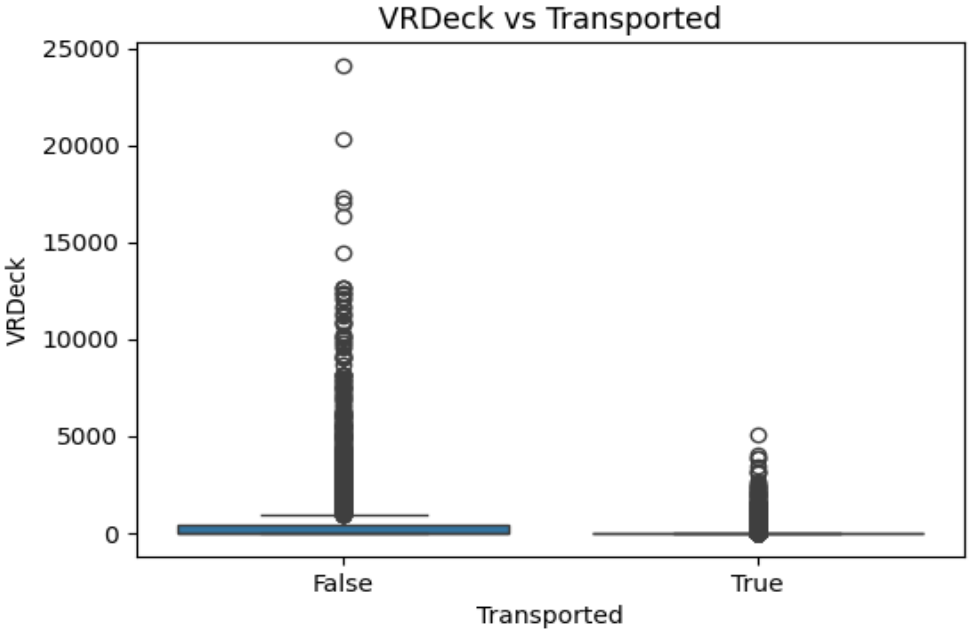





FoodCourt vs Transported
30000
25000
20000
15000
10000
5000
0
FoodCourt
False
True
Transported


ShoppingMall vs Transported
20000
15000
10000
5000
0
ShoppingMall
False
True
Transported


Spa vs Transported
Spa
20000
15000
10000
5000
0
False
True
Transported


VRDeck vs Transported
VRDeck
25000
20000
15000
10000
5000
0
False
True
Transported

Preprocessing Summary

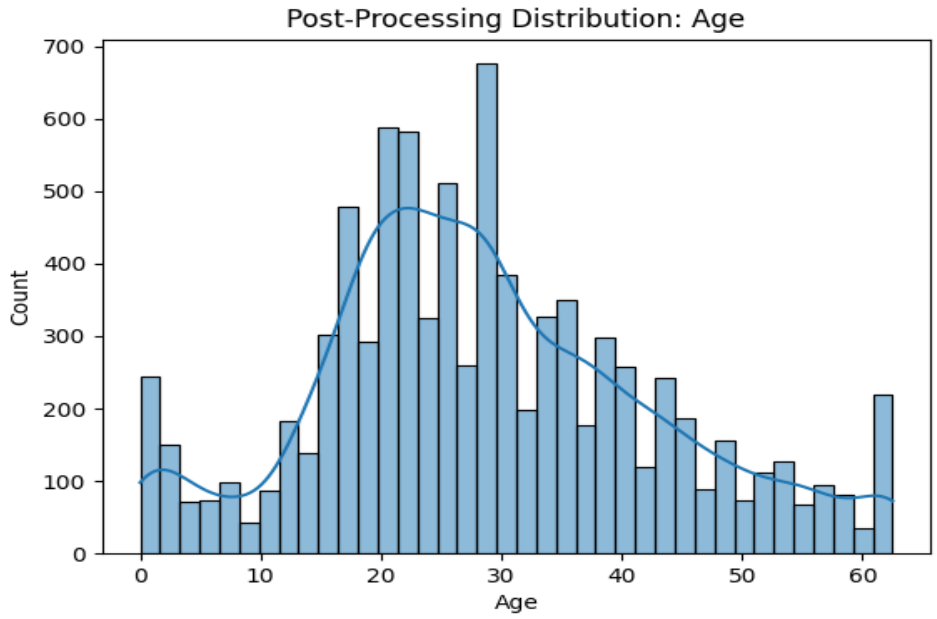
Feature	Missing Count	Missing %	Strategy
HomePlanet	201	2.31	constant
CryoSleep	217	2.5	constant
Cabin	199	2.29	constant
Destination	182	2.09	constant
Age	179	2.06	mean
VIP	203	2.34	constant
RoomService	181	2.08	mean
FoodCourt	183	2.11	mean
ShoppingMall	208	2.39	mean
Spa	183	2.11	mean
VRDeck	188	2.16	mean
Name	200	2.3	constant

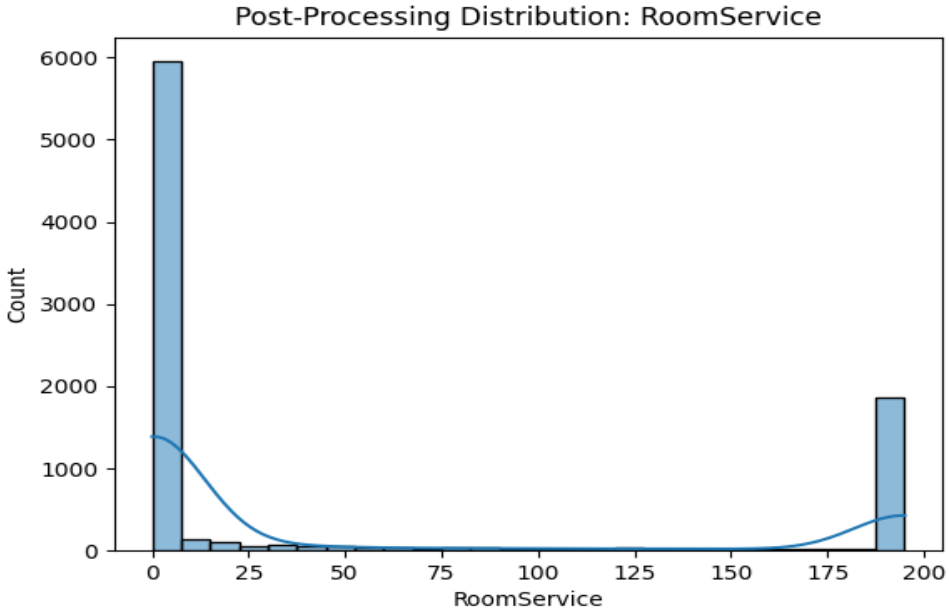
Duplicate rows removed: 0

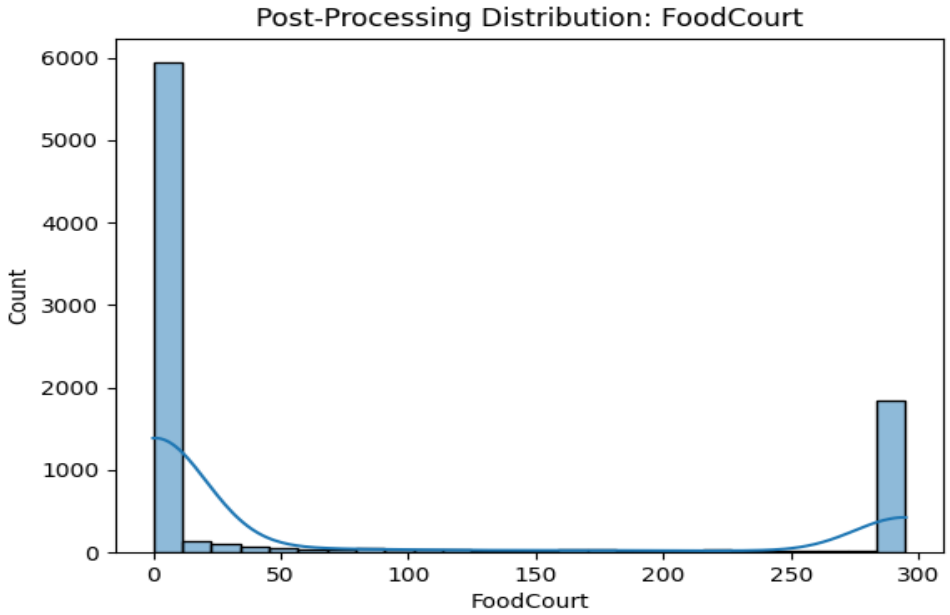
Feature Statistics (After Preprocessing)

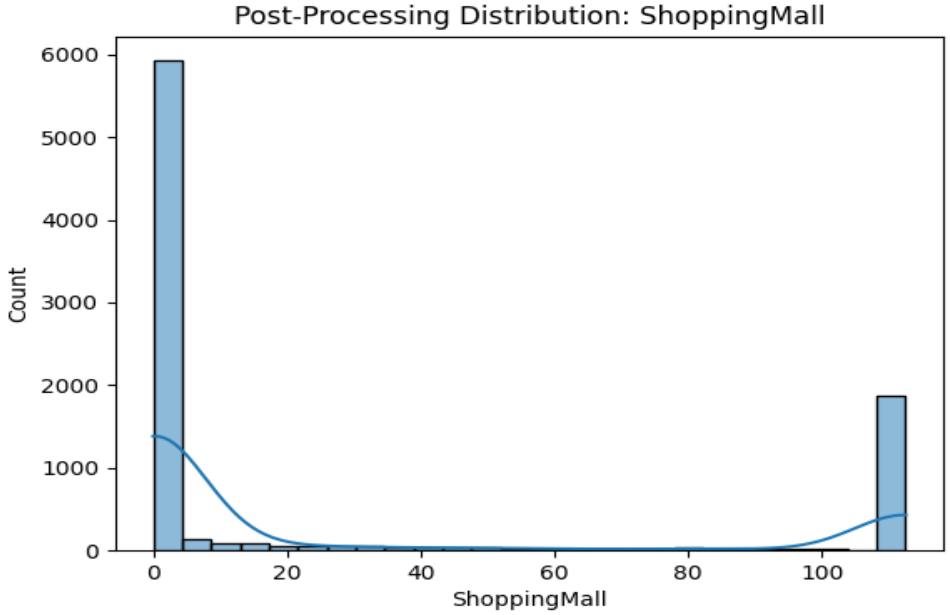
Feature	Type	Unique	Mean	Std	Min	Max
PassengerId	str	8693				
HomePlanet	str	4				
CryoSleep	object	3				
Cabin	str	6561				
Destination	str	4				
Age	float64	65	28.736	14.093	0.0	62.5
VIP	object	3				
RoomService	float64	189	48.543	80.56	0.0	195.0
FoodCourt	float64	267	73.446	121.634	0.0	295.0
ShoppingMall	float64	112	28.069	46.473	0.0	112.5
Spa	float64	214	55.486	91.036	0.0	222.5
VRDeck	float64	173	44.245	73.121	0.0	177.5
Name	str	8474				

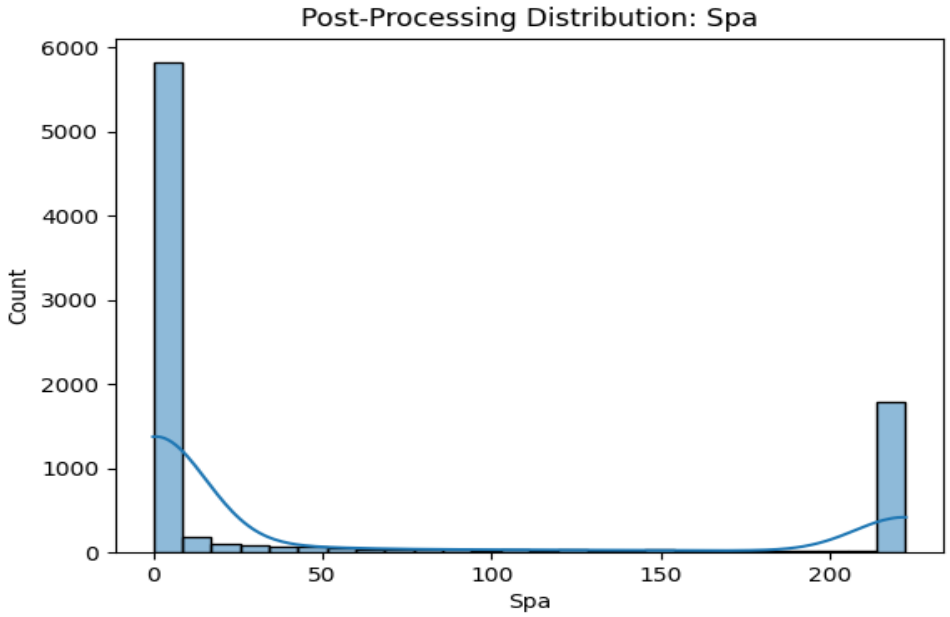
Distributions After Preprocessing

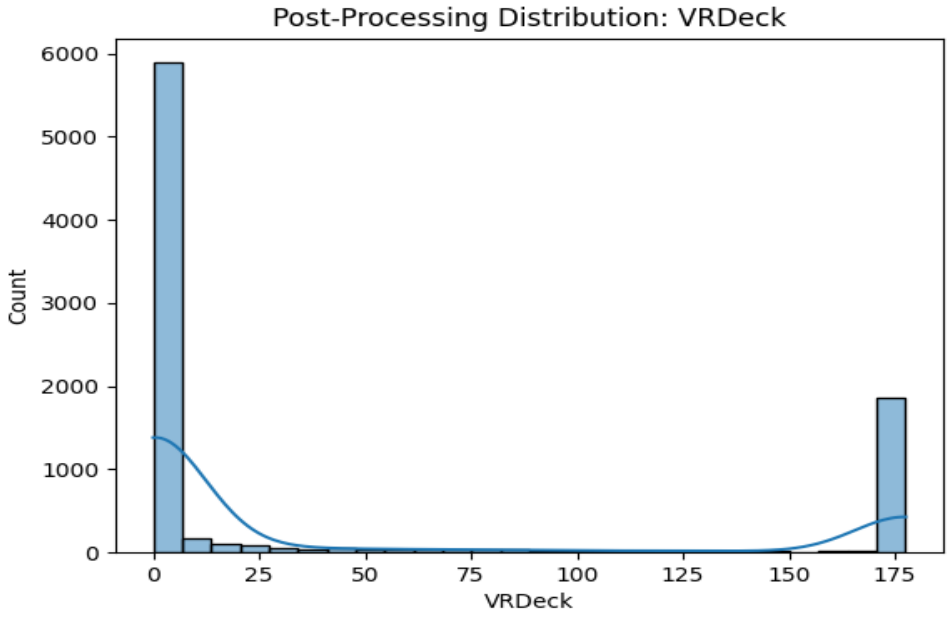





Post-Processing Distribution: FoodCourt
Count
FoodCourt
0
1000
2000
3000
4000
5000
6000
0
50
100
150
200
250
300


Post-Processing Distribution: ShoppingMall
Count
ShoppingMall
0
1000
2000
3000
4000
5000
6000
0
20
40
60
80
100


Post-Processing Distribution: Spa
Count
Spa
0
1000
2000
3000
4000
5000
6000
0
50
100
150
200

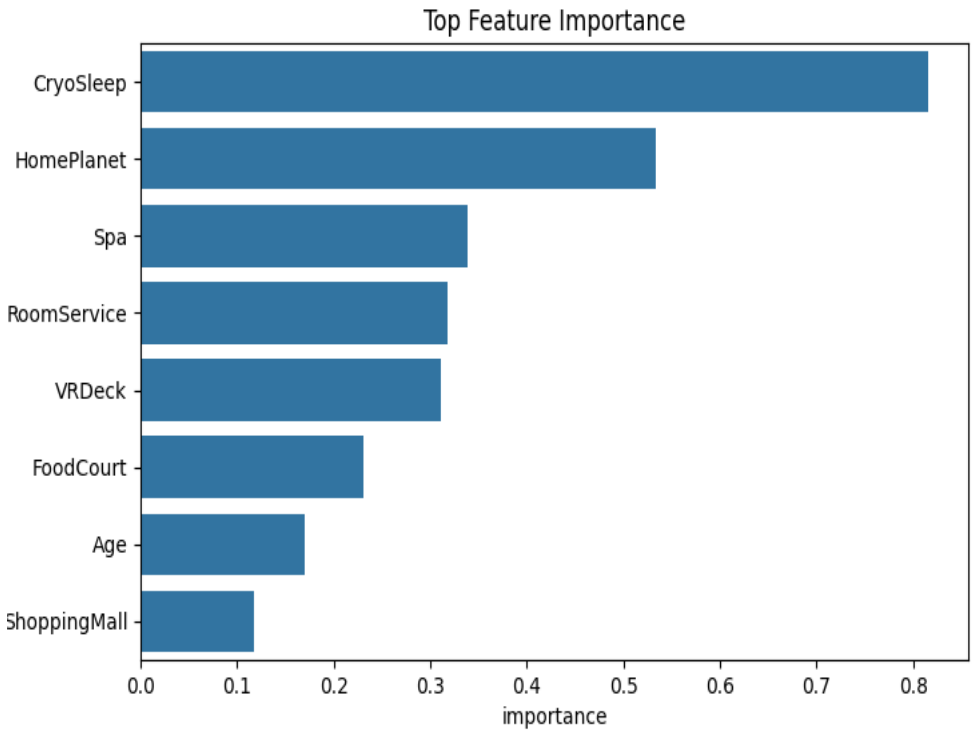

Post-Processing Distribution: VRDeck
Count
VRDeck
0
1000
2000
3000
4000
5000
6000
0
25
50
75
100
125
150
175

Feature Selection

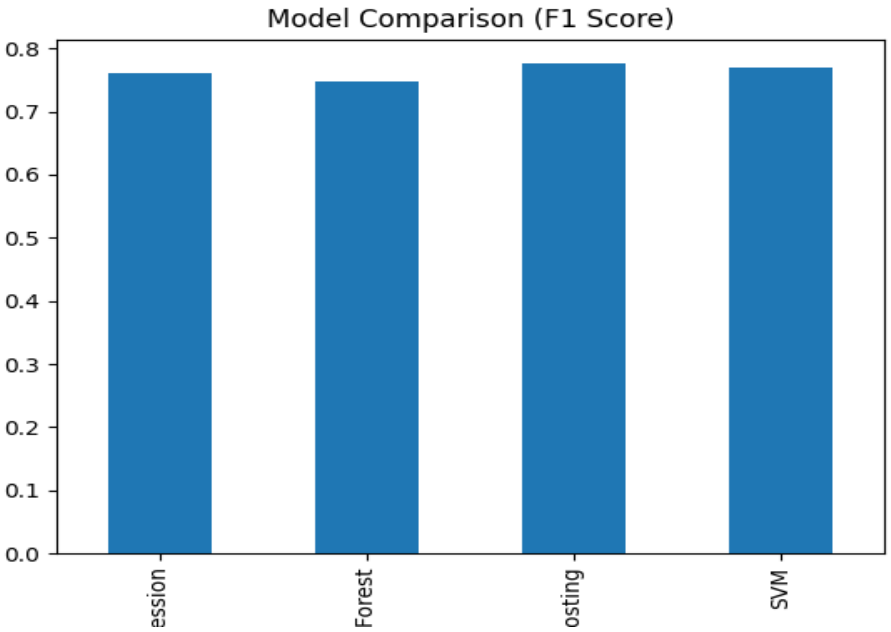
Initial features: 13
Final features: 8

Final Features Used
Age
Spa
RoomService
CryoSleep
VRDeck
FoodCourt
HomePlanet
ShoppingMall

Feature Importance

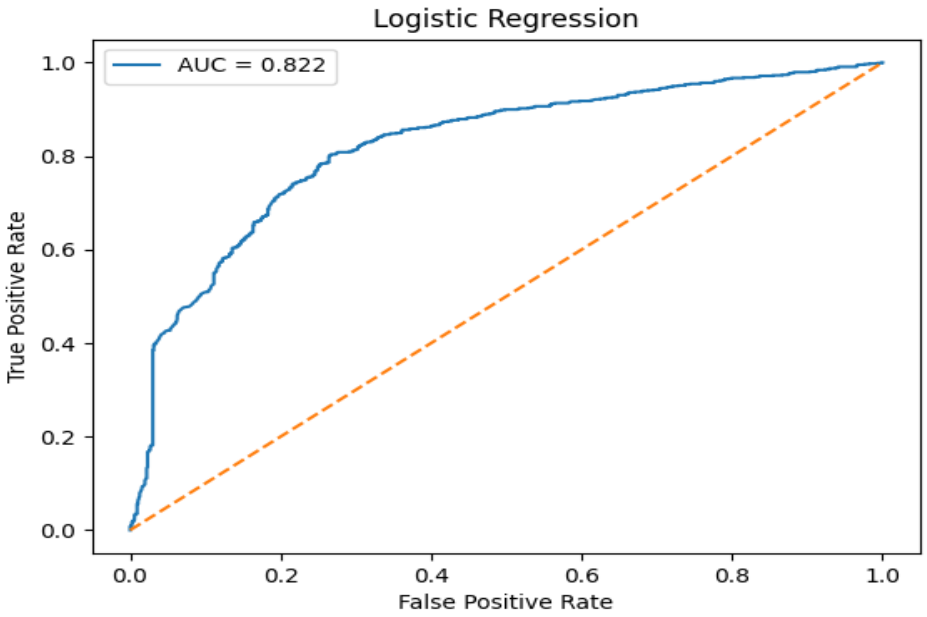


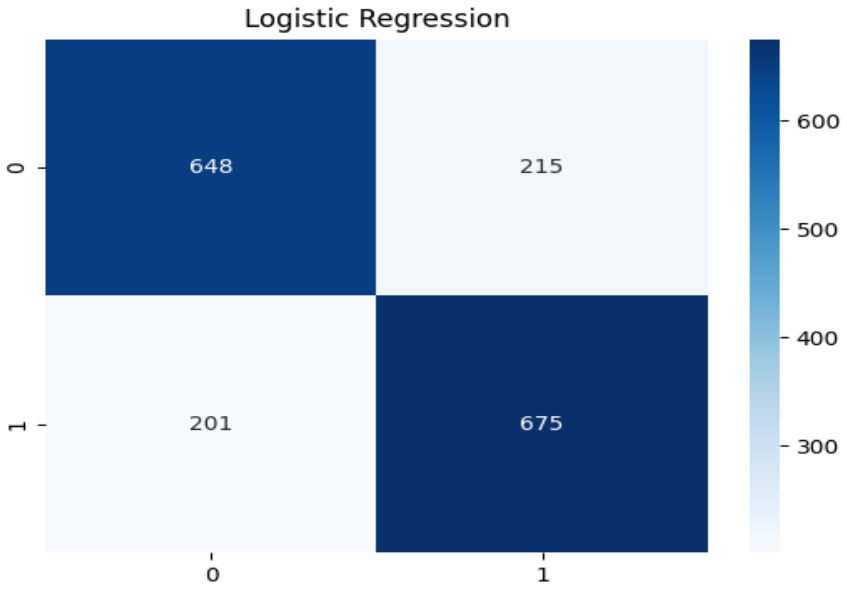
Model Comparison



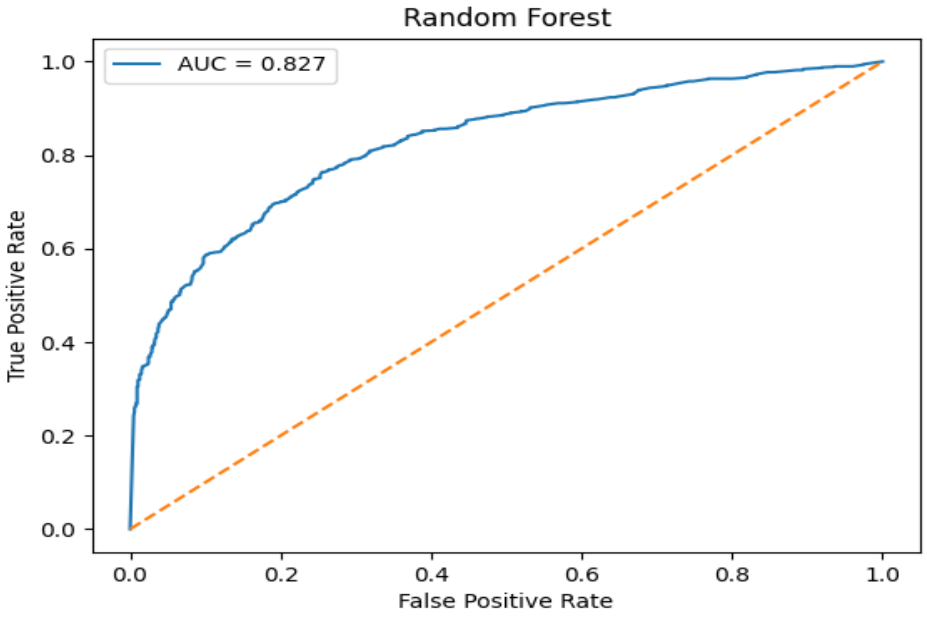
Model Evaluation

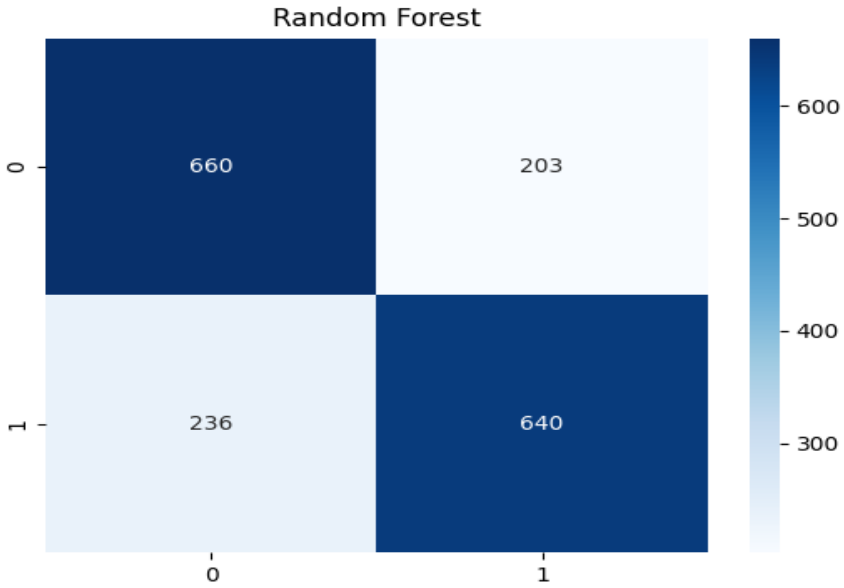
Logistic Regression



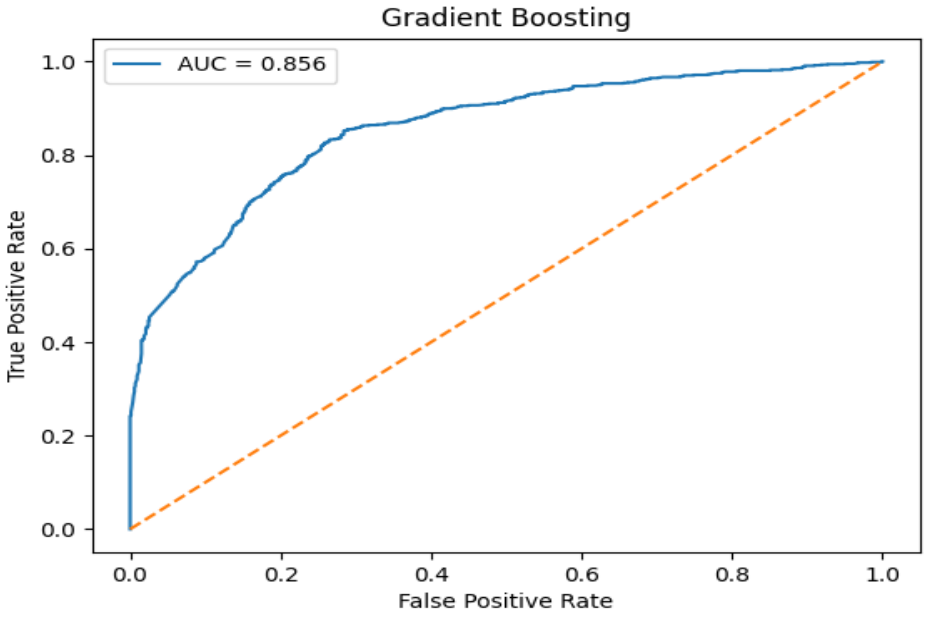


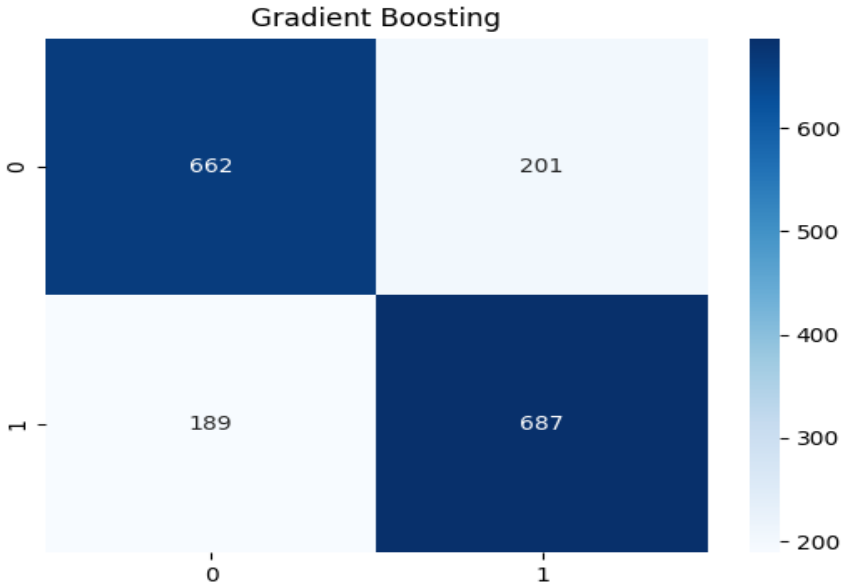
Random Forest



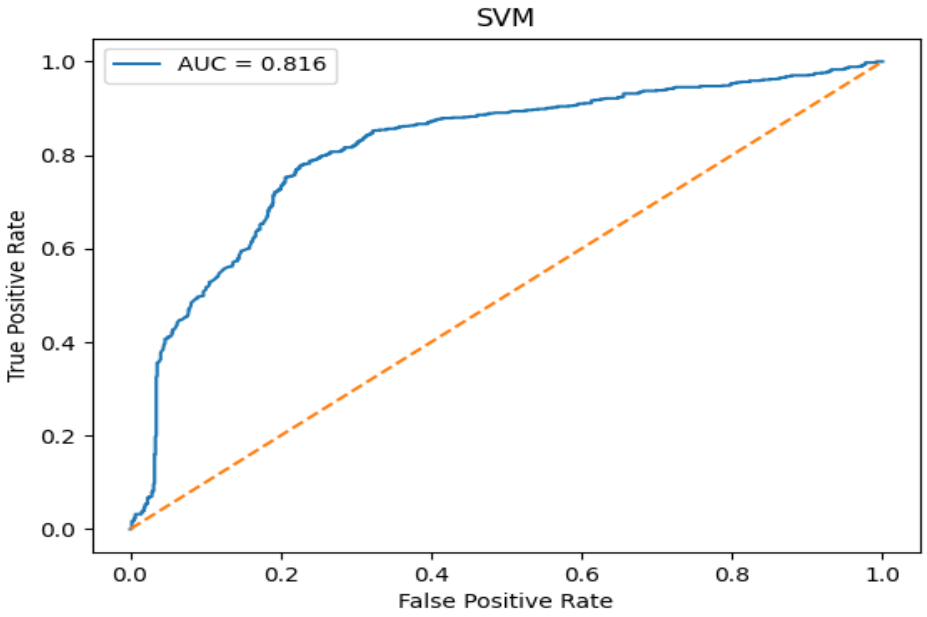


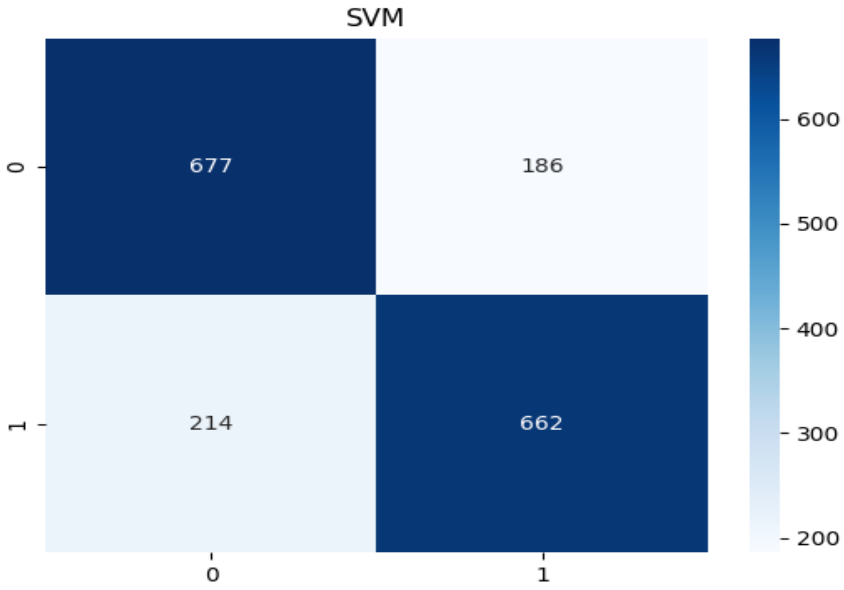
Gradient Boosting


Gradient Boosting
AUC = 0.856
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0


Gradient Boosting
662
201
189
687
0
1
600
500
400
300
200

SVM


SVM
AUC = 0.816
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0


SVM
677
186
214
662
0
1
600
500
400
300
200

Best Model Selected

The AutoML pipeline selected **Gradient Boosting** as the best performing model.

Pipeline Summary

Step	Description
Data Loading	CSV loaded into pandas
Preprocessing	Missing values handled, duplicates removed, outliers capped
Feature Selection	MI filter + correlation removal + RF importance
Model Training	Multiple models trained and evaluated
Best Model Selection	Model with best F1 / R2 selected
Model Explainability	SHAP used for feature importance

Appendix: Execution Logs

```
============================================================
Testing AutoML Pipeline
============================================================

[1/4] Preprocessing...
✓ Data loaded: 8693 rows, 14 columns

Handling Missing Values
-----------------------
  - Filled HomePlanet (201) using constant
  - Filled CryoSleep (217) using constant
  - Filled Cabin (199) using constant
  - Filled Destination (182) using constant
  - Filled Age (179) using mean
  - Filled VIP (203) using constant
  - Filled RoomService (181) using mean
  - Filled FoodCourt (183) using mean
  - Filled ShoppingMall (208) using mean
  - Filled Spa (183) using mean
  - Filled VRDeck (188) using mean
  - Filled Name (200) using constant

Removing Duplicates
-------------------
✓ Removed 0 duplicates

Handling Outliers
-----------------
  - capped 162 outliers in Age
  - capped 1843 outliers in RoomService
  - capped 1828 outliers in FoodCourt
  - capped 1859 outliers in ShoppingMall
  - capped 1785 outliers in Spa
  - capped 1838 outliers in VRDeck

Preprocessing Complete
----------------------
Final shape: (8693, 13)

[2/4] Detecting problem type...
✓ Problem Type: classification

[3/4] Selecting features...

Starting Smart Feature Selection
--------------------------------
Initial Features: 13

Removing Constant Features
--------------------------

Removing ID / High Cardinality
------------------------------
  ✗ PassengerId
  ✗ Cabin
  ✗ Name

Encoding Categorical Features
-----------------------------
  ✓ Encoding HomePlanet
  ✓ Encoding CryoSleep
  ✓ Encoding Destination
  ✓ Encoding VIP

Filtering by MI
---------------
```

```
Calculating Mutual Information
------------------------------

Top MI Features
  1. CryoSleep                          0.1136
  2. Spa                                0.0667
  3. RoomService                        0.0599
  4. VRDeck                             0.0494
  5. FoodCourt                          0.0397
  6. ShoppingMall                       0.0369
  7. HomePlanet                         0.0190
  8. Age                                0.0101
  9. Destination                        0.0042
  10. VIP                                0.0000
  ✗ Destination (low MI)
  ✗ VIP (low MI)

Removing Correlated Features
----------------------------

Model Based Feature Selection
-----------------------------

Top Model Features
  1. Age                                0.2340
  2. Spa                                0.1435
  3. RoomService                        0.1358
  4. CryoSleep                          0.1281
  5. VRDeck                             0.1181
  6. FoodCourt                          0.0863
  7. HomePlanet                         0.0788
  8. ShoppingMall                       0.0755

Feature Selection Complete
--------------------------
Final Features: 8
✓ Features retained: 8/13

[4/4] Training models...

Splitting Data
--------------
✓ Training set: 6954 samples
✓ Test set: 1739 samples
✓ Features: 8

Training Models
---------------
  Training Logistic Regression... ✓
  Training Random Forest... ✓
  Training Gradient Boosting... ✓
  Training SVM... ✓

✓ Trained 4 models

Evaluating Models
-----------------

Logistic Regression:
  Accuracy: 0.7608
  Precision: 0.7608
  Recall: 0.7608
  F1-Score: 0.7608
  ROC-AUC: 0.0000

Random Forest:
  Accuracy: 0.7476
  Precision: 0.7480
  Recall: 0.7476
  F1-Score: 0.7475
```

```
  ROC-AUC: 0.0000

Gradient Boosting:
  Accuracy: 0.7757
  Precision: 0.7758
  Recall: 0.7757
  F1-Score: 0.7757
  ROC-AUC: 0.0000

SVM:
  Accuracy: 0.7700
  Precision: 0.7703
  Recall: 0.7700
  F1-Score: 0.7700
  ROC-AUC: 0.0000

Best Model Selection
--------------------
✓ Best Model: Gradient Boosting
Metrics:
  Accuracy: 0.7757
  Precision: 0.7758
  Recall: 0.7757
  F1-Score: 0.7757
  ROC-AUC: 0.0000

Explaining model using SHAP
---------------------------

 Features
  CryoSleep                     0.8163
  HomePlanet                    0.5342
  Spa                           0.3381
  RoomService                   0.3184
  VRDeck                        0.3115
  FoodCourt                     0.2302
  Age                           0.1697
  ShoppingMall                  0.1176
■ Model + pipeline saved as automl_full.pkl

■ Test completed successfully!
✓ Best Model: Gradient Boosting
✓ Sample Metric: 0.7757

[5/5] Generating AutoML report...
```

B. Feature Selection Summary

Id Removed: PassengerId
High Cardinality Removed: Cabin, Name
Constant Removed: None
Mi Removed: Destination, VIP
Correlation Removed: None
Final Features: Age, Spa, RoomService, CryoSleep, VRDeck, FoodCourt, HomePlanet, ShoppingMall

C. Model Performance Summary

Logistic Regression: Accuracy: 0.7608 | Precision: 0.7608 | Recall: 0.7608 | F1-Score: 0.7608 | ROC-AUC: 0.0000
Random Forest: Accuracy: 0.7476 | Precision: 0.7480 | Recall: 0.7476 | F1-Score: 0.7475 | ROC-AUC: 0.0000
Gradient Boosting: Accuracy: 0.7757 | Precision: 0.7758 | Recall: 0.7757 | F1-Score: 0.7757 | ROC-AUC: 0.0000

SVM: Accuracy: 0.7700 | Precision: 0.7703 | Recall: 0.7700 | F1-Score: 0.7700 | ROC-AUC: 0.0000

D. Final Recommendation

Best Model Selected: **Gradient Boosting**